

CoRoT-2b

R-1674

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_130920 · created 2026-08-02T10:53:16+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456555.7188 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.128 +/- 0.024
Transit depth (Rp/Rs)^2	1.63 +/- 0.61 [%]
Orbital Inclination (inc)	89.27 +/- 1.85
Airmass coefficient 1 (a1)	0.964 +/- 0.021
Airmass coefficient 2 (a2)	0.0191 +/- 0.0059
Scatter in the residuals of the lightcurve fit is	2.17 %
Best Comparison Star	#2 - [127, 301]
Optimal Aperture	3.4
Optimal Annulus	9.52
Transit Duration (day)	0.0917 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.128 $\pm$ 0.024 vs 0.1667 (deviation 1.61 σ)
Duration fit vs published	0.0917 d vs 0.095 d (deviation 0.92 σ)
Mid-time O-C	-5.3 min
Depth SNR	5.3
Residual scatter	2.17 %

Light curve

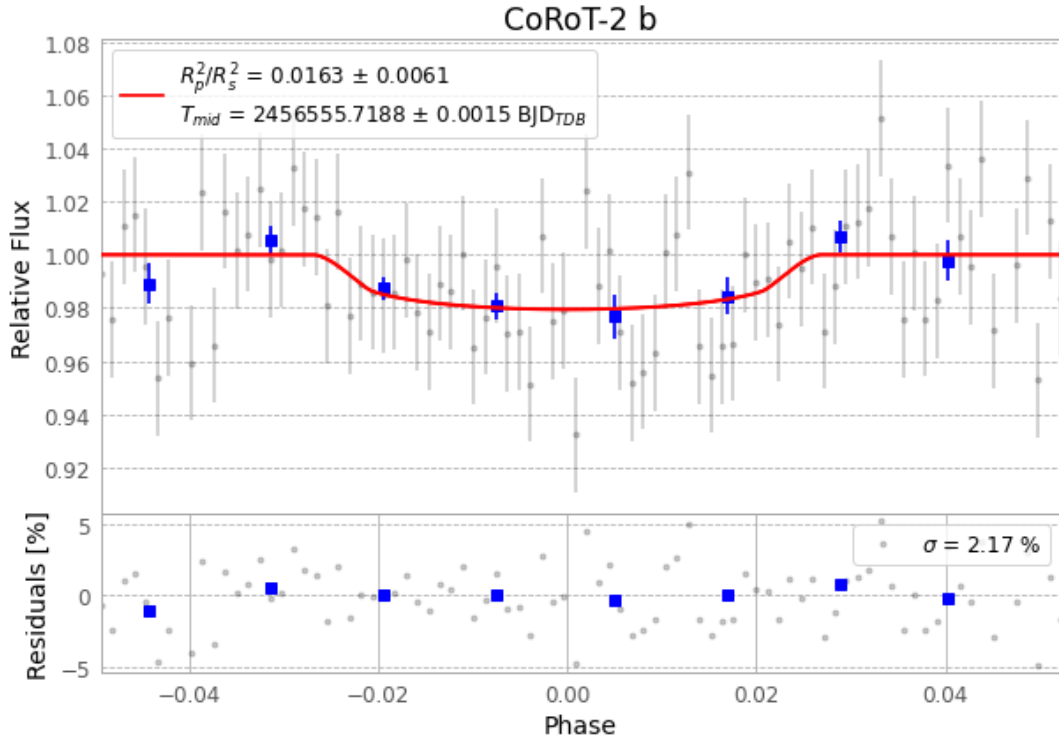


Figure 1 — FinalLightCurve_CoRoT-2 b_2013-09-19.png (SHA-256 bf20c81227d89406...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [217, 276], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 dcf96afc00d260cc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), CoRoT-2130920030612.FITS → CoRoT-2130920072112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-130920.log (bba5e0ef7c0c...), FinalLightCurve_CoRoT-2 b_2013-09-19.png (bf20c81227d8...), FinalLightCurve_CoRoT-2 b_2013-09-19.csv (e8e0ec2ac02a...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 10:53Z.